COURSE DESCRIPTION FORM

INSTITUTION FAST-NUCES

PROGRAM (S) TO BE EVALUATED BS Computer Science/DS

A. Course Description

Course Code	AI-2002
Course Title	Artificial Intelligence
Credit Hours	3 + 1
Prerequisites by Course(s) and Topics	Data Structures
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	**1.** Assignments **(10 %)** **2.** Project **(10 %)** **3.** Quizzes **(10 %)** **4.** Midterms **(30 %)** **5.** Final Exam **(40 %)**
Course Instructor	Umme Ammarah
Office Address	Office # 41 (F block)
Office Hours	**Tuesday-Thursday** 11:30 to 02:30 PM
Course TA	Eesha Waqar (l226780@lhr.nu.edu.pk)
URL (if any)	
Textbook (or **Laboratory Manual** for Laboratory Courses)	1. Stuart Russell, Peter Norvig - Artificial Intelligence A Modern Approach - (3rd Edition) 2. George F. Luger - Artificial Intelligence
Reference Material	1. Tom Mitchell Machine Learning
Course Policies:	• Quizzes may be announced or surprised. No Late Submissions or Makeup Quizzes. • Students bear all the responsibility for protecting their assignments. In case of cheating, both parties will be considered equally responsible. • Plagiarism in any work (labs, quizzes, Assignments, midterms, and final exams) from any source, Internet, or student will result in an F grade or deduction of absolute marks.

Course Goals
Mapping of CLOs & PLOs

CLOs	PLOs
1. Summarize the notions of rational behavior and **intelligent agents.**	2
2. Express appreciation of the goals, subareas, achievements, and difficulties of AI.	2
3. Apply methods of **blind as well as informed search** on the problem related to the programs.	3
4. Summarize concepts and approaches in knowledge representation, planning, learning, robotics, and other AI areas.	2
5. **Program AI applications** by implementing the concepts learnt in the course	4

Topics Covered in the Course, with Number of Lectures on Each Topic
(assume 15-week instruction and one-hour lectures)

Week	Lecture Details
1	– Introduction – Agent and Agent Architectures – Knowledge Representation – PEAS
2-5	**Strategies for State Space Search:** **Blind/Uninformed/Brute-force Search** 1. Depth First Search (DFS) 2. Breadth First Search (BFS) 3. Iterative Deepening Search (IDS) 4. Uniform Cost Search (UCS) Direction of Search, Branching Factor **Heuristic/Informed Search** 1. Best First Search 2. A* Search Informedness, Monotonicity, Admissibility, Optimality, Completeness **Adversarial Search Algorithms (Game Playing)** 1. Minimax Search, a. Alpha-Beta Pruning, b. Evaluation Functions
6-7	**Local Search:** 1. Hill Climbing Search **Evolutionary Search:** Genetic Algorithm 1. Genetic Algorithm Introduction 2. Solution representation 3. Fitness evaluation function

		4. Selection mechanism (Roulette wheel, Rank based, Tournament) 5. Genetic operators (Crossover, Mutation) 6. Replacement method 7. Generation gap, Elitism
	8-10	**Supervised learning** **Regression** – Simple Linear Regression **Machine Learning Introduction, Supervised Learning, Classification (ANN)** **–** Architecture: Feed-Forward Neural Network – Activation Functions (Linear,Sigmoid, Relu, tanh, Softamax) – Training Algorithms: Perceptron and common activation functions, Linear separability of a perceptron – Perceptron Learning Rule – Multilayer Perceptron
	11-12	– Back propagation algorithm (Gradient Descent Learning) – Issues of ANN (Over-fitting, Under-fitting, Validation testing) – Performance measures: Confusion Matrix (Binary & Multi Class) – Evaluation metric: Accuracy, recall. Precision, F1 score
	13	**Supervised learning** - Decision Trees - Convolutional Neural Network (fwd prop only)
	14-15	**Unsupervised learning -** Clustering and its application – K–means algorithm, – K-medoids algorithm – Agglomerative clustering

Laboratory Projects/Experiments Done in the Course	Yes			
Programming Assignments Done in the Course	Yes			
Class Time Spent on (in credit hours)	**Theory**	**Problem Analysis**	**Solution Design**	**Social and Ethical Issues**
	30%	30%	35%	5%
Oral and Written Communications	Every student must submit a Project (In Group) and appear for viva.			

Instructor Name: Umme Ammarah

Instructor Signature _____

Date: 19th January 2026